

BRIAN BELL

Automation Engineer | Robotics Engineer | Manufacturing Engineer | Systems Engineer
Tampa, FL | briantbell.work@gmail.com | briantbell.github.io
U.S. Citizen | No sponsorship required

PROFESSIONAL SUMMARY

Mechanical engineer specializing in automation, robotics, and systems engineering for regulated manufacturing. Leads robotics and electromechanical programs end-to-end, from system architecture through integration, testing, validation, and production delivery. Skilled in Python control software, cobot integration, computer vision, and AS9100 / CMMC compliant manufacturing.

PROFESSIONAL EXPERIENCE

Mechanical Engineer

Oct 2025 – Present

Custom Manufacturing & Engineering

Tampa, FL

- Managed multiple concurrent manufacturing programs end-to-end, frequently recovering distressed programs; owned contract breakdown, sourcing, execution, and reporting through production delivery.
- Authored and interpreted engineering drawings, requirement and test specifications, and GD&T; coordinated with machine shops from prototype through production. Achieved first article approval with zero redlines and sourced prototype chassis fabrication at \$300 against quotes of \$2,500 and \$9,000.
- Led electromechanical system redesign and testing to resolve component obsolescence, lowering manufacturing cost and complexity for core systems while improving technical data in parallel.
- Performed conceptual design and R&D across multiple robotics programs, including modular military payload rover systems spanning locomotion, sensors, and payload integration.
- SolidWorks CAD subject matter expert: created models and technical drawings across programs, reviewed and approved drawings, established CAD standards, and trained team members.
- Subject matter expert for reflow soldering profile development and thermal chamber testing across product lines; leading reinstatement of vibration table testing capability.

Cadet Engineer

Jun 2024 – Dec 2024

MWA Environmental

Brisbane, QLD, Australia

- Built Python automation systems to parse and structure high-volume datasets (200+ weather stations, 43,200+ octave-band noise measurements), enabling filterable analysis and compliance-driven threshold comparison.

ENGINEERING PROJECTS

Autonomous Cobot-Enabled Workcell

Oct 2025 – Present

Custom Manufacturing & Engineering — Technical Lead

Tampa, FL

- Technical lead for a commercial high-precision dual-cobot manufacturing platform supporting tele-operated and autonomous operation. Drove development across system architecture, electromechanical integration, PC-based control software, operator HMI, sensor fusion, OpenCV / ArUco visual servoing, and NVIDIA Isaac Sim digital twin. Interfaced directly with executive leadership and clients; contributed to R&D proposals for continued development.

STOA — Research Knowledge Graph Platform

2026 – Present

Personal Project

Full-Stack Development

- Built a full-stack academic knowledge graph platform: arXiv and PDF ingestion, LLM-assisted concept extraction (Claude API), sentence-transformer embeddings for similarity search, and Neo4j graph storage. Designed the pipeline, schema, and architecture from scratch.

EDUCATION

B.Eng. (Honours), Mechanical Engineering

Jun 2025

Queensland University of Technology — Minors in Computer Science & Finance

Brisbane, Australia

First Class Honours | GPA 7.0 / 7.0 (equivalent to 4.0 / 4.0 U.S.) | Executive Deans' List for Academic Excellence

TECHNICAL SKILLS

Automation, Controls & Robotics: Collaborative Robotics (Cobots), Universal Robots, Robot Integration, Control System Design, HMI Design, Control Software, Sensor Fusion, Computer Vision, Visual Servoing, Test Automation, Validation, ROS2, NVIDIA Isaac Sim, OpenCV

Systems & Mechanical Engineering: Requirements Analysis, System Architecture, Electromechanical Integration, Failure Analysis, Root Cause Analysis, Mechanical Design, Design for Manufacturing (DFM), SolidWorks, CAD, ANSYS, FEA, GD&T, Technical Drawing, Tolerance Analysis, 3D Printing

Programming & Software: Python, C++, SQL, Neo4j (Cypher), TensorFlow, Linux

Manufacturing, Test & Compliance: Reflow Soldering, Thermal Chamber Testing, Vibration Testing, First Article Inspection, AS9100, CMMC, FedRAMP, Aerospace & Defense Manufacturing